Lolla Narasimhachittisaisriuday

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.68
TIRUMALA JUNIOR COLLEGE,RAJAHMUNDRY	Class XII (BIEAP)	2019	10.00%
SRI SIDDHARTHA E.M HIGH SCHOOL, RAVULAPALEM	Class X (BSEAP)	2017	10.00%

EXPERIENCE

Mechanical Intern -- **Abci Infrastructures Pvt. Ltd., Silchar, Assam.**

- Supported workshop operations and coordinated maintenance activities for large-scale construction equipment.
- Assisted teams in inspection, troubleshooting, servicing, and repair of hydraulic and mechanical systems.
- Managed maintenance records, spare-part tracking, job cards, and equipment performance documentation.

Graduate Engineering Trainee -- **Reliance Industries Limited, Jamnagar, Gujarat.**

- Supported operation and maintenance activities of a large-scale cryogenic Air Separation Unit integrated with refinery operations.
- Coordinated maintenance activities involving compressors, turbo-expanders, heat exchangers, and cold boxes.
- Collaborated with operations and maintenance teams to support reliable equipment performance and troubleshooting.

PROJECTS

Multistage Botnet Attack Detection Using Sequence Learning[GitHub]

- **Built a Deep Learning IDPS** using TensorFlow/Keras to detect 4-stage botnet attacks on the CTU-13 dataset.
- **Modeled NetFlow sequence dynamics** with a 30-flow sliding window across stacked LSTM & GRU architectures.
- **Achieved 97.79% accuracy & 97.76% F1-score**, enabling early threat detection within just 2 network flows.

PRISM: Risk-Adaptive Runtime Control for Enterprise AI[GitHub]

- **Architected a model-agnostic control layer** to screen LLM outputs across Quality, Cost, and Responsibility dimensions.
- **Implemented adaptive assurance floors** to prevent 100% of critical failures while reducing deep checks by 30%.
- **Engineered deterministic policy routing** (PASS/REPAIR/VERIFY/ESCALATE) with automated PII redaction and audit logging.

AI Wildlife Detection & Threat Deterrent System[GitHub]

- **Fine-tuned YOLOv11 models** for real-time wildlife detection with >90% accuracy and sub-second inference.
- **Engineered acoustic deterrent triggers** based on real-time perimeter distance estimation to prevent conflicts.
- **Architected an AWS alerting pipeline** (SNS, S3, FastAPI) dispatching emergency incident alerts in <2s.

SKILLS

Programming Languages: Python, SQL

AI & ML: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing

Frameworks & Libraries: TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, Hugging Face, NumPy, Pandas, Matplotlib

Generative AI: RAG, Prompt Engineering, AI Agentic Workflows, LLMs

Engineering & Design: SolidWorks, CATIA, ANSYS, Lotus

Mechanical & Mobility: Vehicle Systems, Mechanical Design, CAD, Steering Systems, Equipment Maintenance

Robotics : UAV Autonomous Navigation, GPS-Denied Navigation, Visual SLAM/VIO, Object Detection, ROS 2

Tools ans Platforms : Git, Github, MySQL, Jupyter notebook, Google Colab

RELEVANT COURSES

Machine Learning & AI: Foundations of Machine Learning, Deep Learning, Random Variables and Stochastic Processes.

Systems & Networking: Advanced Computer Networks, IoT, Autonomous Navigation, Automotive Communication and Sensing

Smart Mobility: Traffic Engineering & Intelligent Transportation Systems, Vehicle Dynamics and Modeling.

POSITIONS OF RESPONSIBILITY

Steering Department Head -- **Team Mech Estremo Racing, NIT AGARTALA.**

- Led the steering department and coordinated design activities for the Baja SAE vehicle.
- Managed steering design, analysis, optimization, and validation using Lotus Shark, SolidWorks, and ANSYS.
- Coordinated technical documentation and presented the department's final report at the SAE M-Baja technical event.

Teaching Assistant -- **Multi-Agent Systems, IIT Hyderabad**

- Teaching Assistant for the **Multi-Agent Systems** course under **Dr. Venkatraman Renganathan**.

SCHOLASTIC ACHIEVEMENTS

- Awarded the **FFE Scholarship** throughout all four years of B.Tech, receiving **40,000 per year**.

- One-to-one mentorship with **M. V. Subrahmanya** from **Trane Technologies**, gained industry guidance on engineering and career development.

HOBBIES

- Fitness & strength training
- Running-3k (4days in week)
- Badminton